20 A water tank of heat capacity 5000 J K^{-1} contains 10 kg of water at 25 °C. What is the time taken to raise the temperature of the water to 45 °C using a heater coil of power of 3.0 kW, given that the specific heat capacity of water is 4200 J kg^{-1} K^{-1} ?

A 61 s **B** 280 s **C** 310 s **D** 610 s